In [1]:
```
import pandas as pd
data = pd.read_csv("Wholesale customers data.csv")
data
```

Out[1]:

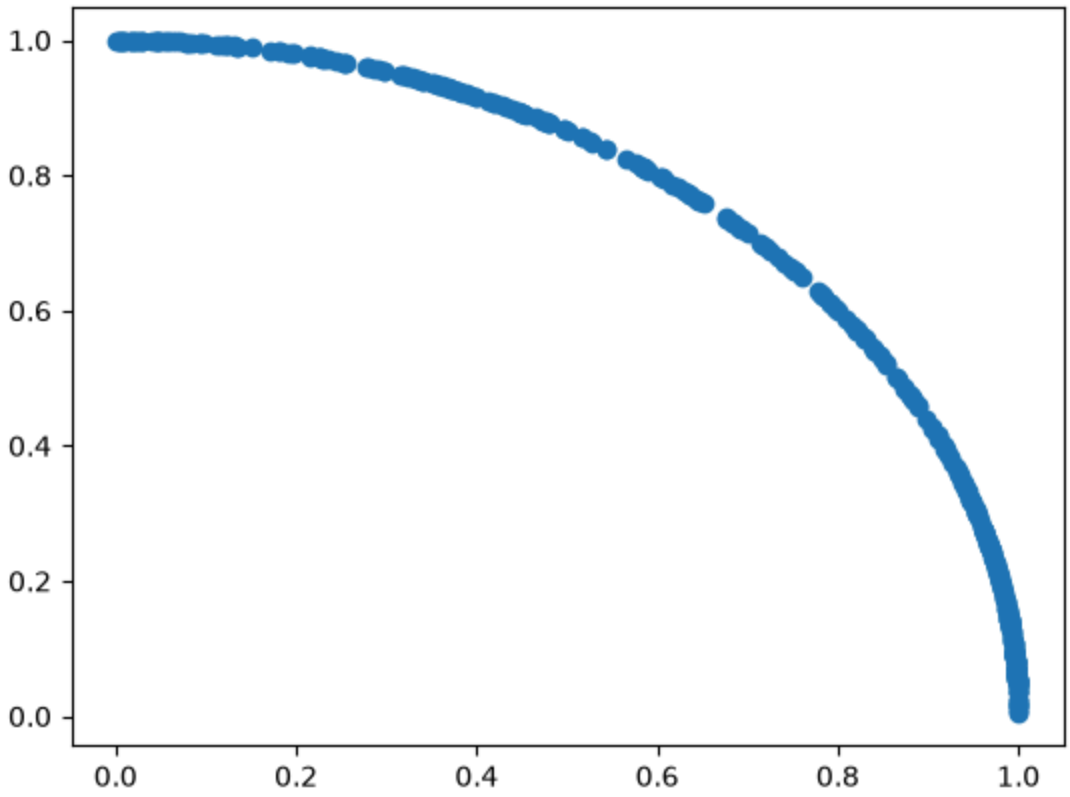
| | Channel | Region | Fresh | Milk | Grocery | Frozen | Detergents_Paper | Delicassen |
|---|---|---|---|---|---|---|---|---|
| **0** | 2 | 3 | 12669 | 9656 | 7561 | 214 | 2674 | 1338 |
| **1** | 2 | 3 | 7057 | 9810 | 9568 | 1762 | 3293 | 1776 |
| **2** | 2 | 3 | 6353 | 8808 | 7684 | 2405 | 3516 | 7844 |
| **3** | 1 | 3 | 13265 | 1196 | 4221 | 6404 | 507 | 1788 |
| **4** | 2 | 3 | 22615 | 5410 | 7198 | 3915 | 1777 | 5185 |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... |
| **435** | 1 | 3 | 29703 | 12051 | 16027 | 13135 | 182 | 2204 |
| **436** | 1 | 3 | 39228 | 1431 | 764 | 4510 | 93 | 2346 |
| **437** | 2 | 3 | 14531 | 15488 | 30243 | 437 | 14841 | 1867 |
| **438** | 1 | 3 | 10290 | 1981 | 2232 | 1038 | 168 | 2125 |
| **439** | 1 | 3 | 2787 | 1698 | 2510 | 65 | 477 | 52 |

440 rows × 8 columns

In [2]:
```
data = data[['Fresh','Milk']]
```

In [5]:
```
from sklearn.preprocessing import Normalizer
scaler = Normalizer()
fitted_data = scaler.fit_transform(data)
```

In [14]:
```
plt.scatter(fitted_data[:,0], fitted_data[:,1])
plt.show()
```




In [6]:
```
fitted_data
```

Out[6]:
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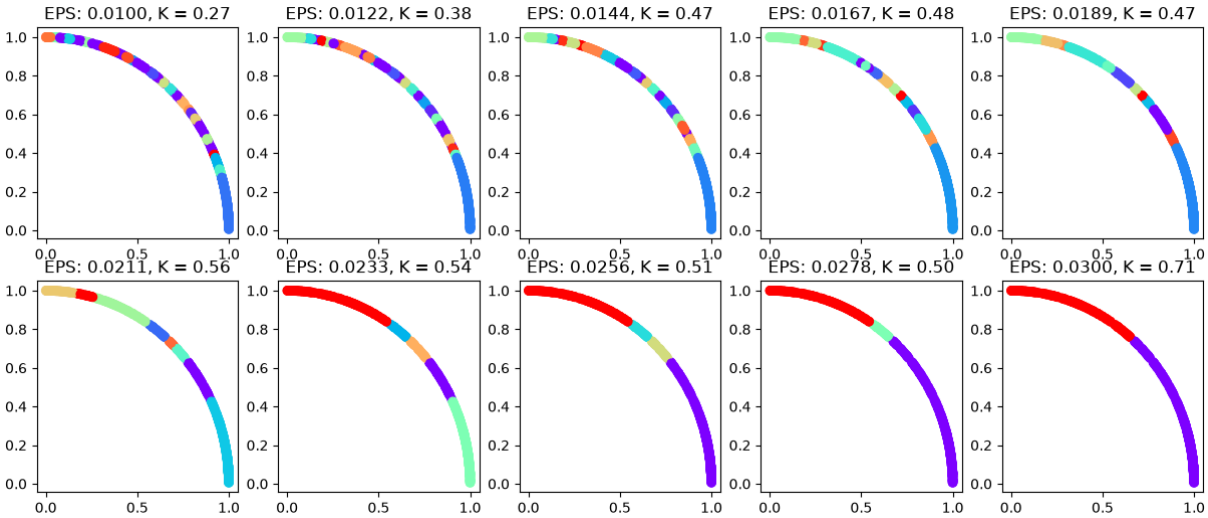
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In [35]:

```
import matplotlib.pyplot as plt
from sklearn.cluster import DBSCAN
from sklearn.metrics import silhouette_score
import numpy as np
ins_num = 10
list_eps = np.linspace(0.01, 0.03, ins_num)
fig, ax = plt.subplots(2, ins_num//2, figsize=(15, 6))
```

```
for idx in range(len(list_eps)):
    cluster = DBSCAN(eps=list_eps[idx], min_samples=5)
    labels = cluster.fit_predict(fitted_data)
    print(f"EPS: {list_eps[idx]}, Number of clusters: {len(set(labels)) - (1
    ax[idx//(ins_num//2), idx%(ins_num//2)].scatter(fitted_data[:, 0], fitte
    silhouette = silhouette_score(fitted_data, labels) if len(set(labels)) >
    if silhouette != -1:
        silhouette = f"{silhouette:.2f}"
    ax[idx//(ins_num//2), idx%(ins_num//2)].set_title(f"EPS: {list_eps[idx]:
plt.show()
```

```
EPS: 0.01, Number of clusters: 20
EPS: 0.012222222222222221, Number of clusters: 18
EPS: 0.014444444444444444, Number of clusters: 17
EPS: 0.016666666666666666, Number of clusters: 15
EPS: 0.018888888888888886, Number of clusters: 12
EPS: 0.02111111111111111, Number of clusters: 8
EPS: 0.02333333333333333, Number of clusters: 5
EPS: 0.025555555555555554, Number of clusters: 4
EPS: 0.027777777777777776, Number of clusters: 3
EPS: 0.03, Number of clusters: 2
```




In [40]:
```
best_eps =  0.027777777777777776
print(f"Best EPS: {best_eps}")
cluster = DBSCAN(eps=best_eps, min_samples=5)
labels = cluster.fit_predict(fitted_data)
cluster.labels_
```

```
Best EPS: 0.027777777777777776
```

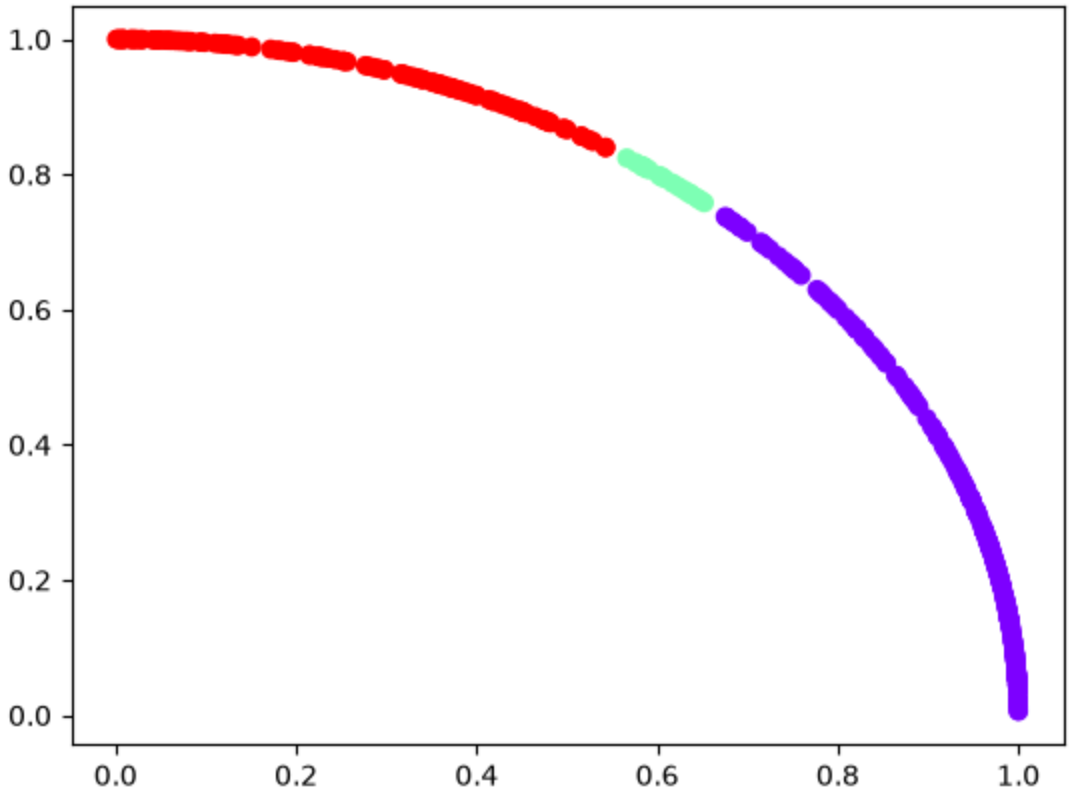
Out[40]:
```
array([0, 1, 1, 0, 0, 0, 0, 0, 0, 2, 2, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0,
       0, 1, 0, 0, 0, 0, 2, 0, 0, 2, 0, 0, 1, 2, 0, 0, 2, 0, 0, 0, 0, 2,
       0, 2, 2, 1, 0, 2, 0, 1, 0, 2, 0, 0, 2, 2, 0, 0, 0, 0, 0, 1, 0, 2,
       2, 0, 2, 0, 0, 0, 1, 0, 0, 0, 2, 0, 0, 2, 0, 2, 0, 0, 0, 2, 2, 0,
       0, 0, 0, 0, 2, 0, 2, 2, 2, 0, 0, 0, 0, 2, 2, 0, 0, 0, 2, 1, 2, 2,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0,
       0, 0, 0, 0, 2, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 2,
       0, 2, 1, 0, 2, 2, 2, 0, 0, 2, 1, 0, 1, 0, 0, 0, 2, 2, 2, 2, 2, 2,
       0, 0, 0, 2, 0, 0, 2, 1, 2, 0, 0, 2, 0, 2, 0, 0, 0, 2, 0, 0, 0, 2,
       0, 0, 2, 2, 0, 1, 0, 2, 0, 2, 2, 2, 0, 2, 0, 2, 2, 2, 2, 0, 2, 0,
       0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 2, 0, 2, 0, 0, 0, 0, 0, 0,
       0, 0, 1, 2, 0, 0, 0, 0, 0, 2, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       2, 2, 2, 0, 0, 0, 0, 2, 2, 0, 2, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 2, 0, 2, 2, 1, 2, 2, 2, 0,
       0, 2, 0, 0, 2, 0, 0, 2, 0, 2, 0, 2, 2, 0, 0, 0, 0, 0, 0, 2, 0, 0,
       0, 1, 0, 0, 0, 0, 0, 0, 2, 0, 2, 2, 2, 2, 0, 2, 2, 0, 0, 2, 0, 2,
       2, 2, 0, 2, 0, 2, 2, 2, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 2, 0, 0, 1, 0, 0, 0, 0, 2, 0, 2, 0, 0, 0, 0, 0, 2, 0, 0, 0,
       2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 2, 2, 0, 0, 0, 2, 0,
       2, 0, 1, 0, 0, 0, 0, 0, 0, 0, 2, 0, 1, 0, 0, 2, 0, 0, 0, 0, 0, 0])
```

In [41]:
```
# Number of noise points
noise_points = np.sum(labels == -1)
print(f"Number of noise points: {noise_points}")
```

```
Number of noise points: 0
```

In [42]:
```
plt.scatter(fitted_data[:,0], fitted_data[:,1], c=labels, cmap='rainbow')
```

Out[42]:
```
<matplotlib.collections.PathCollection at 0x7f09c12d5e50>
```



In [ ]: